

Year 8, Week 5 Lesson 1 - Algebraic expressions	Strategies	Curriculum Specific examples	Active Participation S+A = AP
Set - News article on mis-interpreted signs from around the world http://www.telegraph.co.uk/travel/destinations/asia/china/galleries/Lost-in-translation-hilarious-mistranslated-Chinese-signs/sign7/ - The atom bomb was dropped during WW2 in part because of a mis-translation of the Japanese word 'mokusatsu'	- Use the English language to illustrate the issues that can arise when there are misunderstandings - Concept attainment: Mathematics is no different from English or other spoken languages. If two parties don't understand each other, it will cause problems	Can use examples from the recently written test. Many students probably had the right answer, but miscommunicated it on the test	Group discussion on basic algebraic terms leading to class-wide discussion

Objective & Purpose • Having students treat math as a distinct language with its own rules and terms will help them expand their mathematical vocabulary and assist them in being able to more accurately express themselves in a mathematical context	- Spend 5-10 minutes going over the recently written mini-test - Use examples from the exam to link to the importance of expressing yourself mathematically (ex from tests: students would mix up variables and coefficients, or forget one or the other: - Powerpoint on basic math terms - Khan Academy video https://www.khanacademy.org/math/cc-sixth-grade-math/cc-6th-equivalent-exp/cc-6th-parts-of-expressions/v/expression-terms-factors-and-coefficients - Questions/ Worksheet	- Dwyer Ch. 3 and Nelson Ch. 5.1, 5.2	- Group work, class wide discussion, individual work
Input • most of these terms the students will have heard before, some may be new (coefficient). Getting them to correctly apply them will be new	- Examples will specifically target new language. (Identify the coefficient, variable in an equation. how many terms are there, etc...)		- students will be working with partners, hopefully assisting each other with knowledge gaps
Modelling • While circulating, teachers will notice recurring issues amongst the class and address these as a class • Student examples from recent exam will be used in class (both good and bad)	- At each individual partner grouping as well as class-wide		- Both partner-partner and class-wide discussion

Check for Understanding • Teacher circulation of partner grouping during the mock exam • Class-wide examples on the board • Partner shoulder check • Think-Pair-Share	- Think-Pair-Share - Thumbs up/down - Whiteboard work	- Classwork will include examples from both Dwyer and Nelson	- Students coaching each other and helping with gaps in knowledge - Class-wide discussions on persistent issues
Guided Practice • Examples from textbook and worksheet relevant to powerpoint	- In class Questions	- Questions will come from Dwyer Chapter 3 - Nelson Chapter 5.1,5.2	
Independent Practice • In-class questions/ worksheet plus homework		-	- Take home questions/ worksheet provided by teacher at the end of lesson
Closure • Exit slip asking students to identify the coefficient, variable and constant from a given equation • Emphasize that these rules and skills will be used throughout the year		- Basic algebraic language allows us to communicate effectively during class - Introduction of variables doesn't change rules	